

QUANT RESEARCH MARKET REPORT



Introduction

Quantitative Finance, often referred to as “Quant Finance,” is a discipline that applies mathematical models, statistical techniques, and computational algorithms to analyze financial markets and securities. It encompasses various areas, including risk management, derivative pricing, portfolio optimisation, and algorithmic trading.

The roots of Quantitative Finance trace back to the 1950s with Harry Markowitz’s Modern Portfolio Theory, which introduced the concept of portfolio diversification to optimize returns relative to risk. Subsequent developments, such as the Black-Scholes model for option pricing and advancements in computing power, have significantly expanded the field’s scope.

Key Components

- **Mathematical Modelling:** Utilising stochastic calculus, differential equations, and linear algebra to model financial instruments and market behaviours. E.g., derivatives pricing, risk metrics, portfolio optimisation
- **Statistical Analysis:** Employing statistical methods to identify patterns, correlations, and trends within financial data.
- **Computational Algorithms:** Developing algorithms for tasks such as high-frequency trading, risk assessment, and automated portfolio management.

Applications

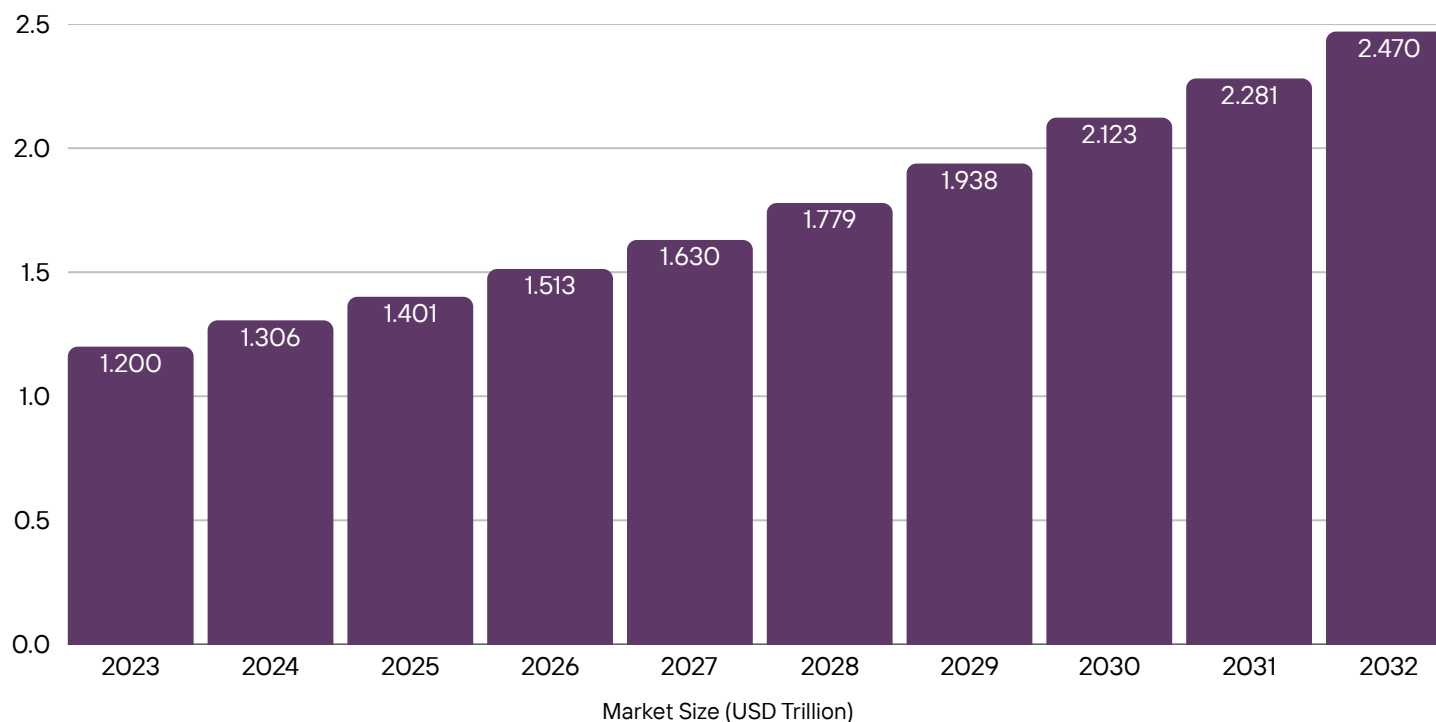
- **Risk Management:** Implementing models like Value-at-Risk (VaR) and stress testing to assess and mitigate financial risks.
- **Derivative Pricing:** Calculating the fair value of complex financial derivatives using models like Black-Scholes and Monte Carlo simulations.
- **Hedge fund management**, where quant funds drove ~22% of global hedge assets and achieved ~8–11% annual returns
- **Algorithmic Trading:** Designing automated trading strategies that execute orders based on predefined criteria and market conditions, with algo-based strategies accounting for ~85% of US equity trading.

- **Portfolio Optimisation:** Allocating assets to maximise returns for a given level of risk, often using techniques like mean-variance optimisation.

Global Market Outlook

Market Size and Growth

As of 2023, the global Quant Fund market is estimated to be valued at approximately USD 1.2 trillion. Projections indicate a Compound Annual Growth Rate (**CAGR**) of **8.5%**, anticipating the market to reach around USD 2.47 trillion by 2032.



Key Drivers

- **Technological Advancements:** The integration of Artificial Intelligence (AI) and Machine Learning (ML) has enhanced the capabilities of quantitative models, allowing for more sophisticated analysis and decision-making.
- **Data Availability:** The proliferation of big data and alternative data sources has provided quantitative analysts with richer datasets for model development.
- **Institutional Adoption:** Institutional investors are increasingly leveraging quantitative strategies to optimise portfolios and manage risks effectively.

Global Trends

- **Machine learning & AI adoption:** up 58% (2019–22), with ML-driven strategies outperforming traditional models by ~20% in 2023.
- **Alternative & ESG data integration:** Used by ~80% of quant firms; ESG-focused strategies saw 25% inflows in 2022.
- **Quantum and blockchain experiments:** Over 60% of institutions exploring quantum computing; blockchain data usage grew 67% (2019–23).
- **DeFi/crypto & digital asset models:** crypto quant strategies and smart-contract-based algos are expanding.
- **Cloud infrastructure & XAI:** heavy cloud use and demand for explainable AI amid tightening regulation.

Quant Finance In India

Market Overview

India's financial landscape is witnessing a gradual yet notable shift towards quantitative methodologies. While traditional investment approaches remain prevalent, the adoption of quantitative strategies is gaining momentum, particularly among institutional investors and fintech startups.

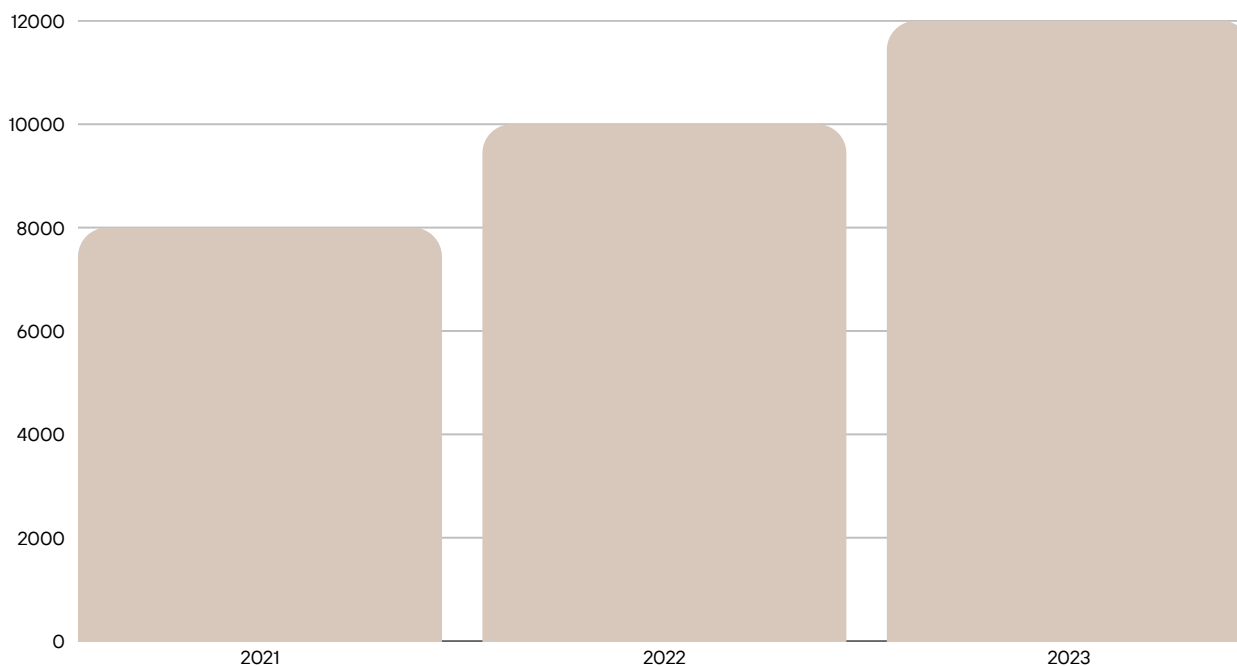
Current Penetration: Quantitative strategies currently account for approximately **1.5%** of the Indian mutual fund market. This is relatively modest compared to markets like the United States, where quant strategies represent over 35% of assets under management.

India's **algorithmic trading market** reached **\$562 million in 2024**, with a projected CAGR of **~9.5%** to reach **\$1.27 billion by 2033**.

India's algorithmic trading is **~0.05% of global quant fund markets**, but its 9–10% CAGR matches global growth.

Growth Trends

- **Assets Under Management (AUM):** Quant-based mutual funds in India manage over **INR 12,000 crore**, reflecting a **50% increase** over the past three years
- **Trading Volume:** Quant strategies contribute to an estimated **25%** of trading volumes in India, up from **15%** in 2019
- **Talent Pool:** India produces around **10,000** new professionals annually with expertise in data science and quantitative Finance, a number projected to grow by **20%** year-on-year



- **Increasing AI/ML platform integration**, infrastructure upgrades (co-location, cloud), and **the rise of HFT**.
- **More educational initiatives** (QuantInsti, IIQF) are driving talent inflow.

Government & Regulatory Stance

- **Algo trading was launched by SEBI in 2008**, enabling DMA, then refined with risk safeguards (kill-switches, audit trails).
- **SEBI sandbox** supports pilot fintech/quant innovations, processing within ~30 working days.
- **Feb 2025 rules**: Retail algo must be tagged, registered via exchange, and audited; algo trading accounted for 96–97% of prop/foreign profit on F&O in FY24.

Significance of Quant in India

Market Overview

- **Mutual funds**: Quant strategies have led to AUM growth (~₹12,000 crore).
- **Asset management & trading**: Automated execution, predictive risk controls, and retail participation have improved efficiency and reduced costs.
- **Fintech & robo-advisors**: Adoption of quant models in platforms like Zerodha, Upstox, and new retail offerings.

- **Proprietary trading firms:** Quant shops (e.g., iRage/QuantInsti tied firms) expanding globally; SEBI is auditing firms like Jane Street's derivatives in India.

Types of Quant Firms

Quantitative finance is implemented across a wide spectrum of financial institutions. These firms differ in their objectives, risk appetite, trading strategies, clients, and regulatory obligations. Below are the major categories:

Asset Management Companies

Asset managers invest on behalf of clients (institutional and retail) across asset classes, increasingly using quantitative models for portfolio optimisation, risk management, and passive strategies.

Global assets under management (AuM) are forecast to reach \$145.4 trillion by 2025, up from \$84.9 trillion in 2016 (6.2% CAGR). Active management is expected to account for \$87.6 trillion (60%), passive management \$36.6 trillion (25%), and alternatives \$21.1 trillion (15%) by 2025

Institutional money managers are responsible for investing funds on behalf of retail and institutional investors. These include mutual funds, pension funds, and ETFs.

Quant Integration:

- AMCs increasingly use quant models to construct portfolios, optimise risk/return tradeoffs, and automate rebalancing.
- Strategies include smart beta, factor investing, and tactical asset allocation.

Global Examples:

- BlackRock, Vanguard, State Street Global Advisors, Invesco QQQ
- These firms manage **trillions in AUM**, often with passive + smart-beta quant overlays.

Indian Examples:

- Nippon India Quant Fund, ICICI Prudential Quant Fund, Tata Quant Fund
- Total AUM in India's quant-based mutual funds crossed **₹12,000 crore (~\$1.45 billion)** by 2024, growing steadily.
- Quant is being used in passive funds, risk-based asset allocation models, and ESG score integration.

Proprietary Trading Firms (Prop Firms)

Prop trading firms use their own capital to trade financial instruments, employing quantitative models for market-making, statistical arbitrage, and high-frequency trading.

Key Features:

- No external clients or AUM obligations.
- Focus on absolute returns and fast execution.
- Heavy investment in low-latency infrastructure, co-location, and custom hardware.
- High secrecy and internal R&D teams to develop edge strategies.

Typical Strategies:

- Statistical arbitrage
- Market-making
- Momentum-based scalping
- Event-driven and latency arbitrage

Global Examples: Jane Street, Jump Trading, DRW, Citadel Securities, IMC, Hudson River Trading

Indian Examples:

- iRageCapital, Alphagrep, Dolat Capital, Samco, Five Star Trading
- India's proprietary trading is heavily involved in **index derivatives (Nifty, Bank Nifty)** and contributes up to **90% of volumes** in certain F&O segments.

Hedge Funds

Hedge funds pool capital from accredited investors to generate returns using a wide range of strategies, with quant hedge funds relying heavily on data-driven, systematic approaches.

The global hedge fund market is projected to grow from \$4,971.75 billion in 2024 to \$5,226.15 billion in 2025 (5.1% CAGR), and is expected to reach \$6,019.79 billion by 2029.

Investment vehicles pooling capital from accredited investors to generate returns using various long/short, directional, or market-neutral strategies—often involving leverage and derivatives.

Key Features:

- May follow both **quantitative** and **discretionary** styles.
- Use of factor models, ML, and statistical systems.
- Performance-based fee structures (commonly 2% management + 20% performance fee).

Quant Hedge Fund Styles:

- Quant Equity (factor-based)

- Global Macro (ML-based forecasting)
- Volatility arbitrage
- Multi-strategy funds combining quant and discretionary layers

Global Examples:

- Renaissance Technologies (Medallion Fund)
- Two Sigma, DE Shaw, Man AHL, AQR Capital
- Renaissance's Medallion Fund has reportedly returned **over 66% annualised (net) between 1988–2018**.

Indian Examples:

- India has limited hedge funds due to regulatory barriers, but firms like Avendus Capital (Alternate Strategies) and Nine Rivers Capital run quant-heavy PMS products.
- Several offshore hedge funds (e.g., Millennium Management) now have India-based quant teams.

Quant Consulting and Data Analytics firms

These firms provide specialised quant research, modelling, training, and data analytics services to hedge funds, banks, brokers, and fintech startups.

Offerings:

- Strategy research
- Portfolio risk diagnostics
- Data engineering for trading platforms
- Quant hiring and talent development
- Algorithm validation and compliance reviews

Global Examples:

- Numerai, WorldQuant, Coremont, Abel Noser
- Platforms like Quantopian (defunct) allowed open-access strategy building by retail quants.

Indian Examples:

- QuantInsti (via EPAT, iRage)
- International Institute of Quantitative Finance (IIQF)
- QuantSense, AlgoBulls provide analytics tools and custom strategies for brokers, HNIs, and traders.

Investment banks

Large global investment banks maintain quant divisions for structuring products, executing trades, risk management, and pricing complex derivatives.

Functions:

- Quantitative research for options pricing, hedging, and fixed income analytics
- Trade execution algorithms (TWAP, VWAP, Implementation Shortfall)
- Risk systems design and stress-testing

Global Examples:

- Goldman Sachs, JP Morgan, Morgan Stanley, UBS, Barclays
- Many employ 1,000+ quants across trading, risk, and structuring teams.

In India:

- While full-scale quant desks are rare, quant roles exist in India-based GICs (Global In-house Centers) of JP Morgan, Morgan Stanley, and Goldman.
- Indian quants are increasingly hired into global roles from campuses like IIT Bombay, ISI, and IIMs.

Fintech Platforms and Robo-advisers

Startups and platforms use algorithmic engines to offer investment advice, trading signals, portfolio construction, and financial planning.

Core Characteristics:

- Scalable, automated, and data-driven
- Serve both retail and institutional clients
- Employ rule-based and ML-backed investment engines

Popular Models:

- SIP optimisation
- Risk scoring and asset recommendation
- Tax-loss harvesting
- Thematic and goal-based investing

Global Examples: Wealthfront, Betterment, SigFig, Acorns

Indian Examples:

- INDMoney, Groww, Zerodha's Smallcase, Kuvera, Scripbox
- Robo-advisors are expected to cross **₹2,500 crore AUM in India** by 2026, with growing retail trust in low-cost, quant-driven advice.

Trends and Innovations in Quantitative Finance

Quantitative Finance is experiencing a transformation driven by technological progress, data availability, and evolving market dynamics. Below are the major trends and innovations shaping the global and Indian quant ecosystem:

High-Frequency Trading (HFT)

HFT refers to ultra-fast, algorithm-driven trading strategies that execute thousands of trades per second, exploiting minute price inefficiencies.

Global Trends:

- Globally, HFT accounts for **50–60% of equity trading volume** in developed markets like the US and Europe.
- Major players include Citadel Securities, Jump Trading, Jane Street, and Virtu Financial.

In India:

- HFT comprises **40–45% of total trades** in Nifty and Bank Nifty futures on NSE.
- Enabled by Direct Market Access (DMA) and co-location services near exchanges, Indian HFT firms (like iRage, Alphagrep) are growing rapidly.
- SEBI has introduced latency audits and mandatory risk management tools (e.g., kill switches) to mitigate systemic risk.

AI/ML-Based Trading and Prediction Models

Overview: Machine Learning (ML) and Deep Learning (DL) models now enhance alpha generation, regime detection, volatility forecasting, and sentiment analysis.

Applications:

- **Supervised learning** (Random Forest, XGBoost) for price prediction.
- **Unsupervised learning** (clustering, PCA) for factor discovery.
- **Reinforcement Learning** (e.g., Deep Q-Learning) for trade execution and portfolio rebalancing.
- **Natural Language Processing (NLP)** is used in news analytics, social media signals (Twitter, Reddit), and earnings call analysis.

Impact:

- AI-driven funds like Two Sigma and Renaissance Technologies have outperformed traditional hedge funds.
- In India, quant teams at Zerodha, small fintech firms, and prop desks are adopting ML for signal generation and risk assessment.

Crypto Quant Strategies (including DeFi Subset)

Rise of Crypto Quants:

- Quant strategies now play a significant role in the **\$1.8 trillion** global crypto market.
- Strategies include statistical arbitrage, mean reversion, liquidity mining, and basis trading.

DeFi Subset:

- Decentralised Finance (DeFi) allows algorithmic strategies across DEXs (e.g., Uniswap, Curve) without intermediaries.
- Strategies include flash loan arbitrage, impermanent loss hedging, and yield optimisation.
- Protocols like **dYdX** and **GMX** are attracting quant developers with on-chain APIs and low-latency execution.

India Scene:

- While crypto regulations are tight, Indian quants participate via offshore accounts or work with international firms.
- India's **WazirX** and **CoinDCX** are exploring ML-backed analytics tools, though still in infancy.

No-Code Quant & Strategy Builders

No-code platforms allow traders and researchers to build, backtest, and deploy strategies without writing code.

Popular Global Tools:

- **QuantConnect**, **Kavout**, **AlgoTrader**, and **Trality**.
- Drag-and-drop interfaces combined with broker integration (Interactive Brokers, Alpaca, Zerodha API).

India-Specific:

- **QuantPower**, **Streak (by Zerodha)**, and **AlgoBulls** are leading the no-code algo revolution.
- These platforms have enabled non-programmers (e.g., retail investors, finance students) to engage in live algorithmic trading with risk checks and SEBI compliance.

Automated Portfolio Management & Robo-Advisors

Robo-advisors use rule-based or ML-enhanced algorithms to provide financial planning, portfolio allocation, and rebalancing without human intervention.

Global Scene:

- Vanguard, Betterment, and Wealthfront collectively manage over **\$500 billion** in AUM.
- Portfolios are constructed using Modern Portfolio Theory (MPT), Risk Parity, and Factor Investing.

In India:

- **INDMoney**, **Groww**, and **Smallcase** are leading robo-advisor and semi-automated portfolio platforms.
- They combine traditional advice with quant models for SIPs, ETF allocations, and tax harvesting.
- A 2024 IAMAI report estimated **robo-managed AUM in India to surpass ₹1,200 crore**, growing at over **30% CAGR**.

Trends and Innovations in Quantitative Finance

Open-Source Tools & Platforms

- **QuantLib**, **Backtrader**, **Zipline**, **PyAlgoTrade**, **TensorTrade**, etc.
- Platforms like **QuantInsti's EPAT** provide structured development.

Datasets & APIs

- **Quandl**, **Bloomberg**, **Reuters**, and exchanges' data feeds.
- Indian brokers/fintech firms now offer **data & model APIs**, with SEBI-approved sandboxes.

Model Validation & Risk Management

- Emphasis on backtesting, stress-testing, live vs simulated performance.
- SEBI mandates risk controls: kill-switches, pre-trade risk checks, algo tagging, and audit trails.